24V 1.4MHz 3A Fast-PWM Synchronous Step-Down Converter

DESCRIPTION

The MT8693 is a fully integrated high efficiency synchronous step-down converter which requires minimum number of external components. It offers very compact solution with up to 3A continuous output current over a wide input range.

The MT8693 employs proprietary Constant On-Time (COT) control scheme providing superior transient response and maintaining constant switching frequency under the continuous conduction mode operation. The internal ramp compensation network allows stable operation with ultra-low equivalent series resistance (ESR) output ceramic capacitors without using external compensation network. An error amplifier in the control loop provides excellent line and load regulation.

The MT8693 integrates extensive protection functions include: UVLO, OCP, UVP and thermal shutdown. The converter is available in a small SOT23_6L package.

FEATURES

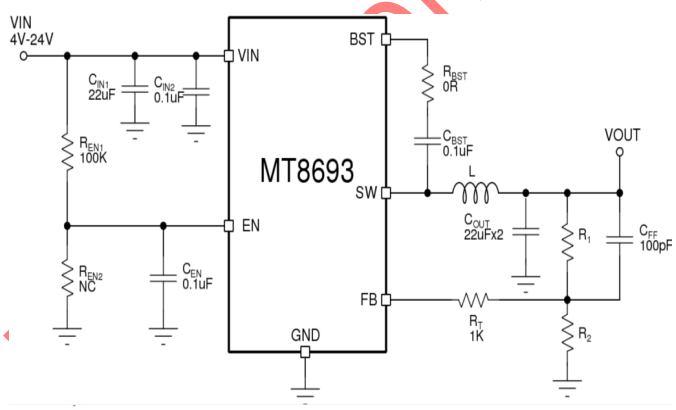
- Input Voltage Range from 4V to 24V
- Output Voltage Adjustable from 0.6V to 24V
- 3A continuous output current
- Support 100% duty cycle Low Dropout Operation
- Stable operation with output low ESR ceramic capacitors
- Fast PWM COT control with superior transient performance
- 1.4MHz Switching Frequency
- Internal 1.2ms Soft-Start
- Integrated 70mΩ/ 47mΩ HS/LS Power Switches
- Accurate EN UVLO threshold
- Thermal Shutdown with Auto recovery.
- Hiccup mode at short circuit protection
- Available in a SOT23_6L Package

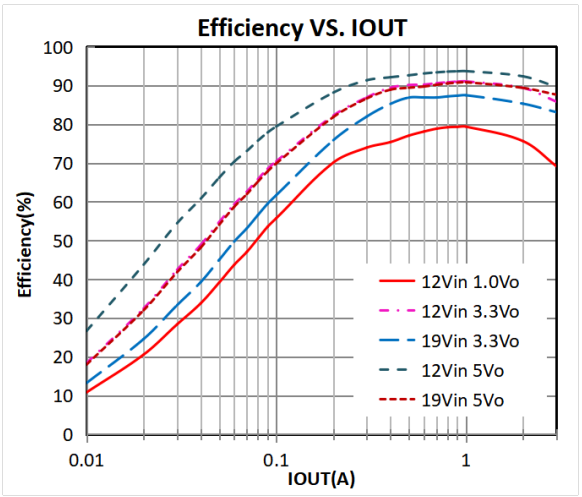
APPLICATIONS

- Laptop Computer
- Tablet PC
- Networking Systems
- Personal Video Recorders
- Flat Panel Television and Monitors
- Distributed Power Systems

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TYPICAL APPLICATIONS



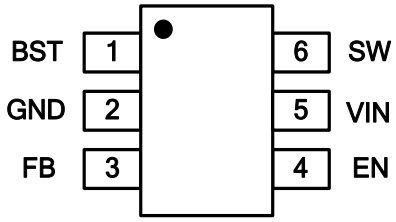


ORDERING INFORMATION

Part No.	Marking	Temp. Range	Package	MOQ
MT8693ASCR	8693A YWWXX	-40°C ~85°C	SOT23_6L	3000/Tape & Reel

Note: Y: Year, WW: Week, XX: Control Code

PIN CONFIGURATION



TOP VIEW

PIN DESCRIPTION

SOT23_6L PIN NO.	SYMBOL	DESCRIPTION
1	BST	Bootstrap. A 100nF ceramic capacitor connected between SW and BST pins is required to form a floating supply for the high-side switch driver.
2	GND	Power ground
3	FB	Feedback. An external resistor divider from the output to GND, tapped to the FB pin, sets the output voltage
4	EN	Enable pin. MT8693 is shut down when this pin is low and active when this pin is high. The hysteretic enable threshold voltage is 1.21V going up and 1.11V going down. Connect EN with VIN through a pull-up resistor or a resistive voltage divider for automatic startup. An external resistor divider from VIN can be used to program a VIN threshold below to stop the MT8693 operation. There is an internal 1MΩ (typical) pull down resistor from EN to AGND.
5	VIN	Supply Voltage. The VIN pin supplies power for internal MOSFET and regulator. The MT8693 operates from a 4V to 24V input rail. An input capacitor is needed to decouple the input rail.
6	SW	Switch Output. Connect this pin to the inductor and bootstrap capacitor. SW node should be kept small on the PCB for good performance and low EMI.

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FUNCTIONAL BLOCK DIAGRAM

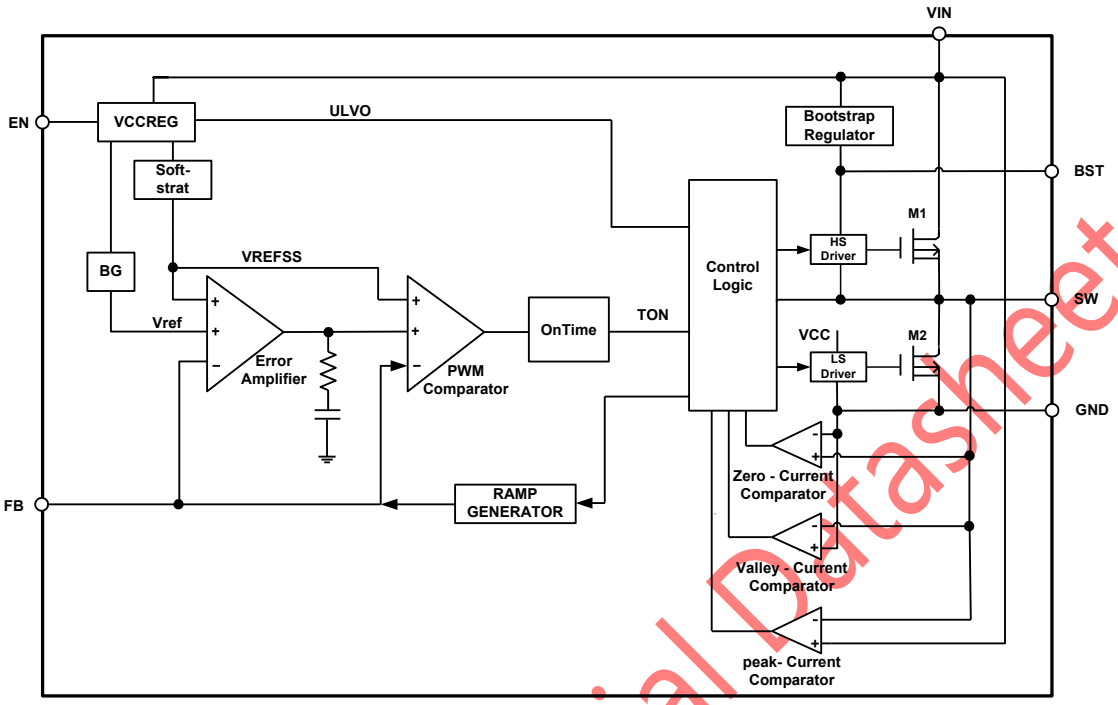


Figure 1—Functional Block Diagram

Absolute Maximum Ratings (Reference to GND) (1)

VIN Voltage,EN .....+30V
SW Voltage ..... –0.3V to V_{IN}+0.3V
Dynamic V_{SW} in 10ns Duration..... –3V to V_{IN}+3V
BS-SW Voltage .....+6V
FB Voltage ..... +6V
Junction Temperature Range ..... -40ºC to+150°C
Storage Temperature Range ..... -65ºC to +150°C
Lead Temperature (Soldering 10s) .....+260°C

Recommend Operating Conditions (2)

Input Voltage (V_{IN}).....+4V to +24V
Operating Temperature Range ..... -40°C to +85°C
Junction Temperature Range, T_J ..... -40°C to 125°C

Thermal Information (3) (4)

Maximum Power Dissipation(T_A=+25°C) 1.02W
Thermal Resistance (θ_{JA}) ..... 122°C/W
Thermal Resistance (θ_{JC}) ..... 81°C/W

Note(1): Stress exceeding those listed “Absolute Maximum Ratings” may damage the device.
Note(2): The device is not guaranteed to function outside of the recommended operating conditions.
Note(3): Measured on JESD51-7, 4-Layer PCB.
Note(4):The maximum allowable power dissipation is a function of the maximum junction temperature T_{J_MAX}, the junction to ambient thermal resistance θ_{JA}, and the ambient temperature T_A. The maximum allowable continuous power dissipation at any ambient temperature is calculated by P_{D_MAX}= (T_{J_MAX}-T_A)/θ_{JA}. Exceeding the maximum allowable power dissipation will cause excessive die temperature, and the regulator will go into thermal shutdown. Internal thermal shutdown circuitry protects the device from permanent damage.

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ELECTRICAL CHARACTERISTICS

T_A = +25°C, V_{IN}=12V and V_{EN}=5V, unless otherwise noted. Typical values are at V_{IN} = 12V, V_{EN} = 5V and V_{OUT} = 5V.

PARAMETER	SYMBOL	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Input Voltage Range	V_{IN}		4		24	V
Shutdown Current	$I_{SHUTDOWN}$	V_{EN} = 0V, V_{IN} = 12V		3		μA
Input Under Voltage Lockout Threshold	V_{UVLO}	V_{IN} Increasing	3.5	3.7	3.9	V
Input Under Voltage Lockout Hysteresis	V_{UVHYS}			300		mV
Supply Current	I_{IN}	V_{FB} = 0.7V, MT8693A		600	1000	μA
Feedback Regulation Voltage	V_{FBREF}		592	600	608	mV
Feedback Current	I_{FB}	VFB=0.6V		10	50	nA
Internal Soft-start Time	T_{SS}			1.2		msec
Switching Frequency	F_{SW}	I_{OUT} = 1A		1.4		MHz
Minimum Off Time (5)	T_{OFF_MIN}			140		ns
Maximum Duty Cycle(6)	D_{MAX}			100		%
HS Main Switch-On Resistance RON_{HS}	HS_{RDS-ON}			70		mΩ
HS Switch Leakage Current	HS_{SWLKG}	V_{IN} = 28V, V_{EN} = V_{SW} =0V		0.1	10	μA
Low Side(LS) Switch Valley Current Limit	I_{VALLEY}			4.4		A
LS Switch Negative Current Limit	I_{NEG}	MT8693A		-1.5		A
LS Switch-On Resistance RON_{LS}	LS_{RDS-ON}			47		mΩ
LS Switch Leakage Current	LS_{SW_LKG}	V_{IN} = V_{SW} = 28V, V_{EN} = 0V		0.1	10	μA
EN On Threshold		V_{EN} ramp up	1.11	1.21	1.31	V
EN Off Threshold		V_{EN} ramp down	1.01	1.11	1.21	V
EN Internal Pull-Down Resistor	R_{EN}		700	1000	1300	kΩ
Thermal Shutdown				160		°C
Thermal Shutdown Hysteresis				30		°C

Note(5): Guaranteed by design, no production test.

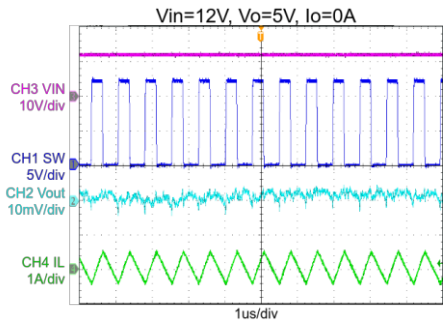
Note(6): When the input voltage approaches the output voltage, the MT8693 will extend the on-time and force the main high side switch remaining on for multiple cycles (>10 μsec). High side switch is only turned off momentarily, and low side switch is forced on shortly (typical 120ns) to refresh the BST capacitor. High side switch will resume on after the BST refresh.

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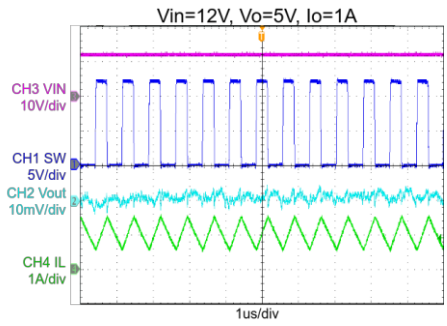
TYPICAL PERFORMANCE CHARACTERISTICS

Performance waveforms are tested on the evaluation board of the Design Example section.
VIN=12V, VOUT = 5V(L=2.2μH), TJ=+25°C, unless otherwise noted.

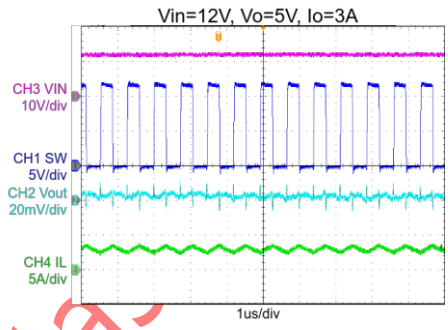
Steady State Test



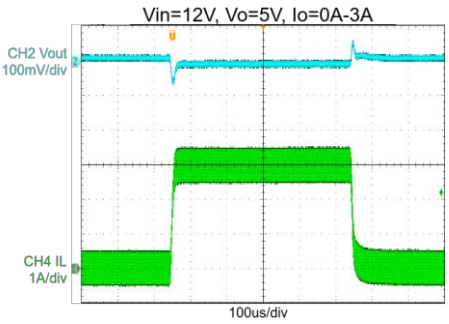
Steady State Test



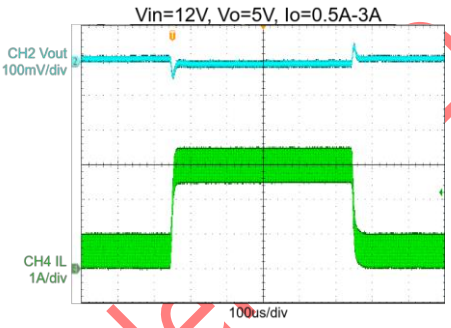
Steady State Test



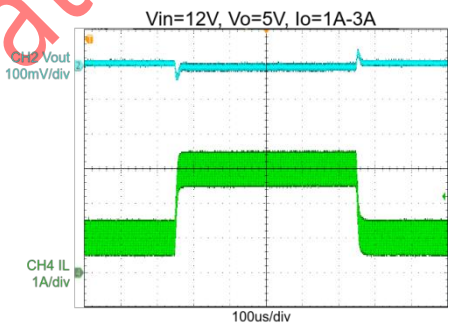
Load Transient Response



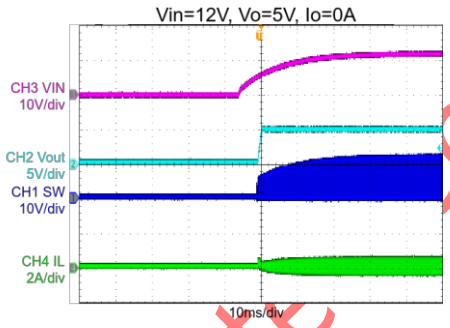
Load Transient Response



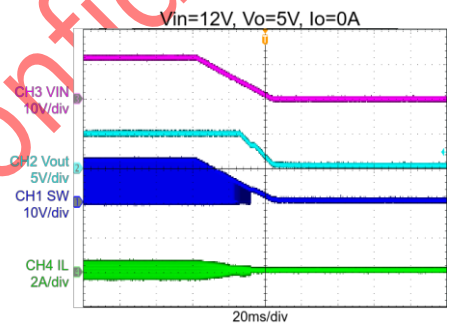
Load Transient Response



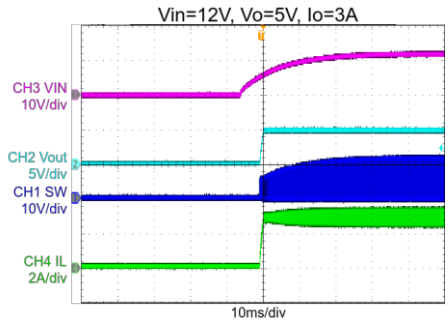
VIN power on



VIN power off



VIN power on

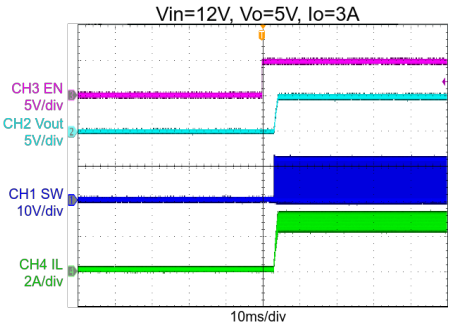


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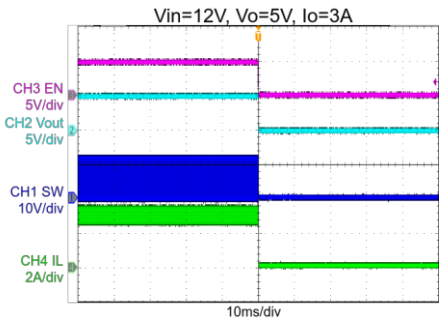
TYPICAL PERFORMANCE CHARACTERISTICS

Performance waveforms are tested on the evaluation board of the Design Example section.
VIN=12V, VOUT = 5V(L=2.2μH), TJ=+25°C, unless otherwise noted.

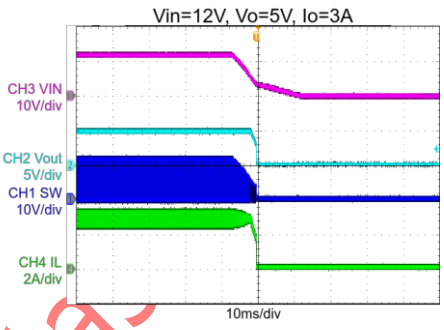
EN power on



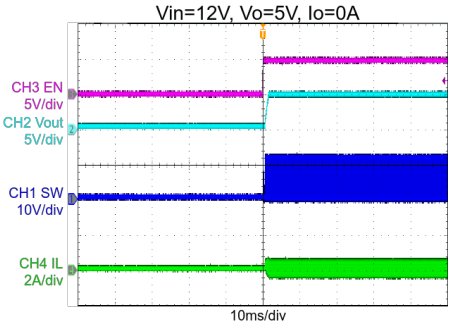
EN power off



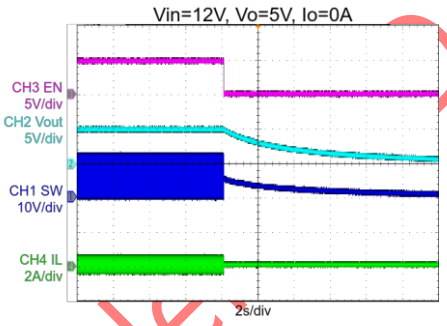
VIN power off



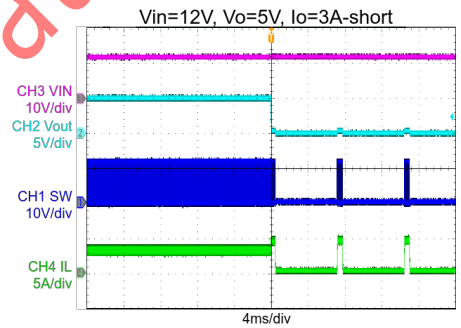
EN power on



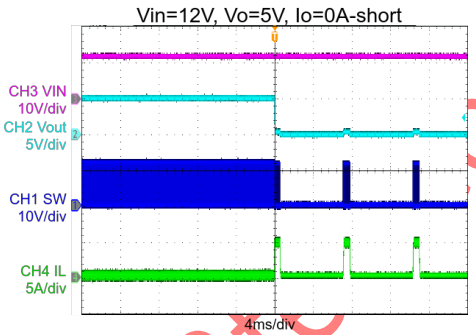
EN power off



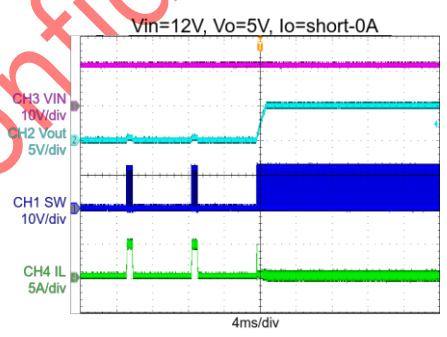
Short Circuit Protection



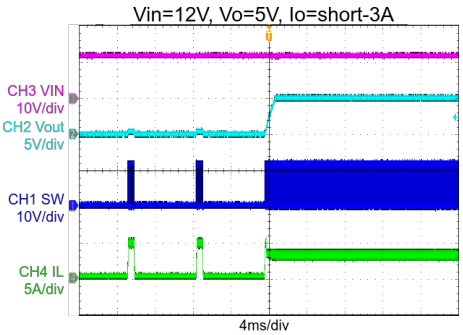
Short Circuit Protection



Short Circuit Protection Recovery



Short Circuit Protection Recovery



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OPERATION

The MT8693 is a fully integrated synchronous step-down converter employing constant on-time (COT) control scheme to achieve superior transient performance. Its proprietary internal ramp compensation offers stable operation with lower ESR ceramic output capacitors without using external complex compensation networks.

Constant On-time Control

The constant on-time control (COT) operates by comparing the feedback voltage V_{FB} with the reference voltage (V_{FBREG}). When FB drops below the reference, the control circuit turns on HS switch immediately for a pre-determined period of time (on-time) to ramp up the inductor current. When this on-time times out, the LS switch is then turned on to ramp down the inductor current. The LS switch is turned off when inductor current triggers negative current limit I_{NEG} or HS switch is turned on again for the next cycle. This operation repeats itself if FB drops below reference again.

The MT8693 uses a proprietary algorithm to calculate the on-time based on input voltage, output voltage and load current to achieve nearly constant switching frequency over entire continuous conduction load current range. The on-time can be estimated as:

$$T_{ON} = \frac{V_{OUT}}{V_{IN}} \times \frac{1}{F_{SW}}$$

Due to its immediate response on FB voltage droop and simplified loop compensation, The MT8693 offers superior transient response compare to traditional fixed frequency PWM control converters.

100% Duty Cycle Low Dropout Operation

When input voltage approaches the output voltage, the MT8693 will extend the on-time toward the maximum on-time to satisfy the duty cycle requirement to regulate the output voltage. If the input further drops to equal or lower than the output level, the MT8693 forces the main high side (HS) switch to remain on for more than one cycle, eventually reaching 100% duty cycle. The 100% duty cycle operation allows the converter to effectively pass through the input voltage directly to output with minimum voltage drops on the HS switch and the inductor. In the low dropout operation mode, the MT8693 turns on HS switch for multiple switching cycles until it turns off HS switch momentarily and turns on low side (LS) switch (typical 120ns) to refresh the BST supply voltage. The LS switch is turned off after the BST refresh pulse, then the HS switch resumes on for multiple switching cycles which gives the effective 100% duty cycle. The refresh BST pulse is needed to charge the BST capacitor and ensure the HS switch driver circuits proper operation.

Enable

MT8693 offers an accurate enable threshold of EN pin, which is typically 1.21V rise and 1.11V fall. The MT8693 is enabled by pulling up the EN pin above1.21V and MT8693 is disabled by pulling down the EN pin above below 1.11V.

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When using the EN pin threshold voltage to program the input startup voltage level, the following equation shall be used:

$$V_{IN-START} = 1.21V \times \frac{R_{UP} + R_{DOWN}//1M\Omega}{R_{DOWN}//1M\Omega}$$

Where the 1MΩ is the internal pull-down resistor on EN pin.

When EN is pulled high, MT8693 will start up if V_{IN} is higher than UVLO threshold. When EN is pulled low, MT8693 will go into shutdown. Tie EN pin to V_{IN} if the shutdown feature is not used.

Soft Start

MT8693 has built-in internal soft start of 1.2 msec. During the soft start period, output voltage is ramped up linearly to the regulation level, independent of the load current and output capacitor value.

Current Limit and Hiccup Mode

The MT8693 has built-in cycle-by-cycle current limit protection to prevent inductor current from running away in any fault conditions. The MT8693 continuously monitors the inductor valley current during its operation. Once the valley current exceeds the limit level, MT8693 will turn on LS and wait for the inductor current to drop down to a pre-determined level before the HS can be turned on again. If this current limit condition is repeated for a sustained long period of time, MT8693 will enter hiccup mode, where it stop switching for a pre-determined period of time before automatically re-try to start up again. It always starts up with soft-start to limit inrush current and avoid output overshoot.

When MT8693 enters valley current limit mode, the peak current is also limit due to the fixed on-time of the HS, and this peak current can be estimated as:

$$I_{PEAK} = I_{VALLEY} + T_{ON} \times \frac{V_{IN} - V_{OUT}}{L}$$

APPLICATION INFORMATION

Setting the Output Voltage

External feedback resistors are used to set the output voltage. 1% resistors are recommended to maintain output voltage accuracy. Refer to typical application circuit on page1, the top feedback resistor R_1 has some impact on the loop stability, the bottom feedback resistor R_2 is recommended 5kΩ~30kΩ, the top feedback resistor R1 can be calculated as:

$$R_1 = R_2 \times (\frac{V_{OUT}}{0.6} - 1)$$

Inductor

The inductor is necessary to supply constant current to the output load while being driven by the switched input voltage. A larger-value inductor will result in less ripple current that will result in lower output ripple voltage. However, a larger-value inductor will have a larger physical footprint, higher series resistance, and/or lower saturation current. A good rule for determining the inductance value is to design the peak-to-peak ripple current in the inductor to be in the range of 30% to

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40% of the maximum output current, and that the peak inductor current is below the maximum switch current limit. The inductance value can be calculated by:

$$L = \frac{V_{OUT}}{F_{SW} \times \Delta I_L} \times (1 - \frac{V_{OUT}}{V_{IN}})$$

Where ΔIL is the peak-to-peak inductor ripple current.

To avoid overheating and poor efficiency, an inductor must be chosen with an RMS current rating that is greater than the maximum expected output load of the application. In addition, the saturation current (typically labeled ISAT) rating of the inductor must be higher than the maximum load current plus 1/2 of in inductor ripple current.

The peak inductor current can be calculated by:

$$I_{L_PEAK} = I_{OUT} + \frac{V_{OUT}}{2F_{SW} \times L} \times (1 - \frac{V_{OUT}}{V_{IN}})$$

Input Capacitor

The input current to the step-down converter is discontinuous and therefore requires a capacitor to supply the AC current to the step-down converter while maintaining the DC input voltage. Ceramic capacitors are recommended for best performance and should be placed as close to the VIN pin as possible. Capacitors with X5R and X7R ceramic dielectrics are recommended because they are fairly stable with temperature fluctuations. The capacitors must also have a ripple current rating greater than the maximum input ripple current of the converter. The input ripple current can be estimated as follows:

$$I_{CIN} = I_{OUT} \times \sqrt{\frac{V_{OUT}}{V_{IN}} \times (1 - \frac{V_{OUT}}{V_{IN}})}$$

The worst-case condition occurs at $V_{IN} = 2V_{OUT}$, where:

$$I_{CIN} = \frac{I_{OUT}}{2}$$

For simplification, choose the input capacitor with an RMS current rating greater than half of the maximum load current. The input capacitance value determines the input voltage ripple of the converter. If there is an input voltage ripple requirement in the system, choose the input capacitor that meets the specification. The input voltage ripple can be estimated as follows:

$$\Delta V_{IN} = \frac{I_{OUT}}{F_{SW} \times C_{IN}} \times \frac{V_{OUT}}{V_{IN}} \times (1 - \frac{V_{OUT}}{V_{IN}})$$

Under worst-case conditions where $V_{IN} = 2*V_{OUT}$:

$$\Delta V_{IN} = \frac{1}{4} \times \frac{I_{OUT}}{F_{SW} \times C_{IN}}$$

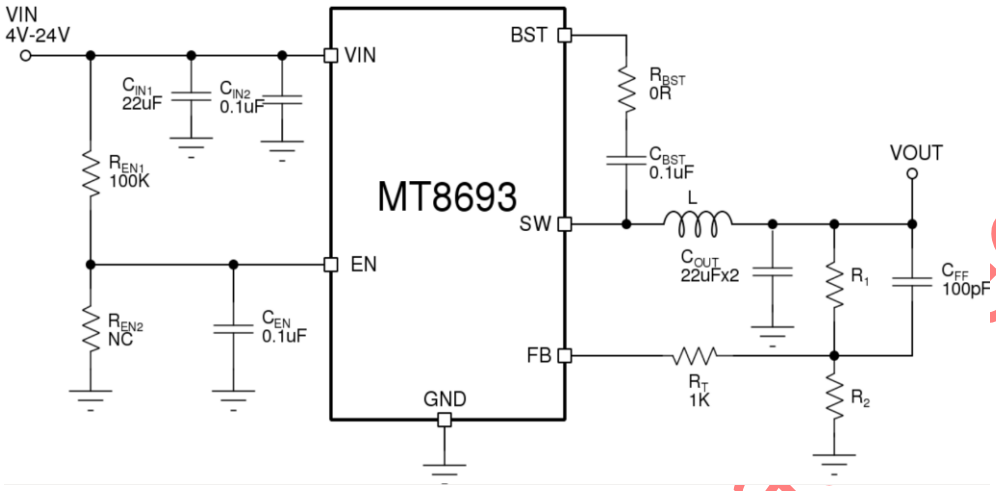
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Output Capacitor

The output capacitor has two essential functions. Along with the inductor, it filters the square wave generated by the MT8693 to produce the DC output. In this role it determines the output ripple, thus low impedance at the switching frequency is important. The second function is to store energy in order to satisfy transient loads and stabilize the MT8693's control loop. X5R or X7R type ceramic capacitors have very low equivalent series resistance (ESR) and provide low output ripple and good transient response. Transient performance can be improved with a higher value output capacitor and the addition of a feed-forward capacitor placed between V_{OUT} and FB. Increasing the output capacitance will also decrease the output voltage ripple. A lower value of output capacitor can be used to save space and cost, but transient performance will suffer and may cause loop instability. When choosing a capacitor, special attention should be given to the data sheet to calculate the effective capacitance under the relevant operating conditions of voltage bias and temperature. A physically larger capacitor or one with a higher voltage rating may be required.

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Application Schematic



EVB BOM List

Qty	Ref	Value		Description	Package
1	C_{IN1}	22μF		Ceramic Capacitor, 35V, X5R	0805
2	C_{IN2}, C_{EN}	0.1μF		Ceramic Capacitor, 50V, X5R	0603
2	C_{OUT}	22μF		Ceramic Capacitor, 35V, X5R	0805
1	C_{BST}	0.1uF		Ceramic Capacitor, 50V, X5R	0603
1	L	Vout=5V	2.2μH	Inductor, Isat> 6A	SMD
		Vout=3.3V	1.5μH		
		Vout=1.0V	1μH		
1	R_1	Vout=5V	150KΩ	Resistor, ±1%	0603
		Vout=3.3V	91KΩ		
		Vout=1.0V	15KΩ		
1	R_2	Vout=5V	20.5KΩ	Resistor, ±1%	0603
		Vout=3.3V	20.5KΩ		
		Vout=1.0V	20.5KΩ		
1	C_{FF}	100pF		Ceramic Capacitor, 50V, X5R	0603
2	R_{EN1}	100KΩ		Resistor, ±5%	0603
1	R_{BST}	0Ω		Resistor, ±1%	0603
1	R_T	1KΩ		Resistor, ±1%	0603
1	Power IC	MT8693		Step-Down DC/DC Converter	SOT23_6L

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PCB Layout Recommendation

(1) The high current paths (GND, VIN, and SW) should be placed very close to the device with short, direct and wide traces.

(2) Put the input capacitors as close to the VIN and GND pins as possible.

(3) Put the decoupling capacitor as close to the VIN and GND pins as possible. Place the Cap close to GND if the distance is long. And place >3 vias if via is required to reduce the leakage inductance.

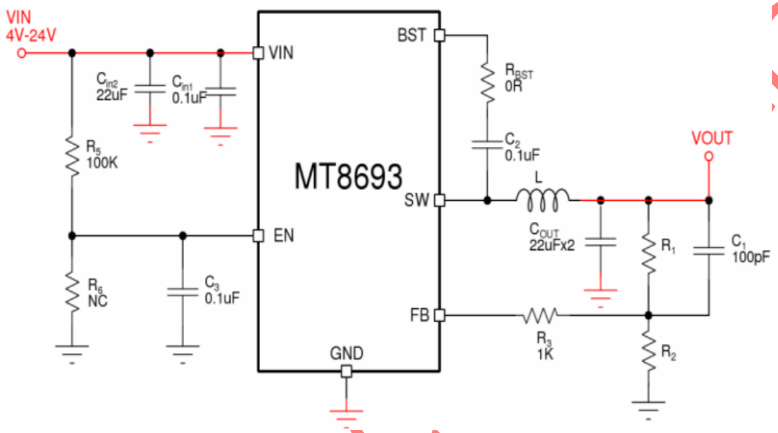
(4) Keep the switching node SW short and away from the feedback network.

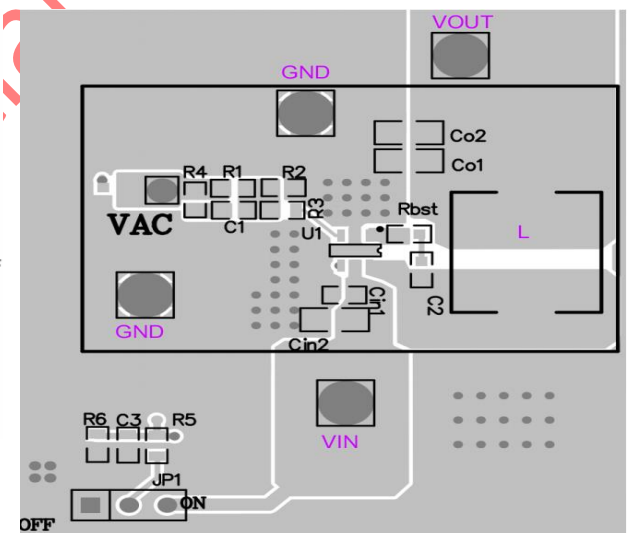
(5) The external feedback resistors should be placed next to the FB pin. Make sure that there is no via on the FB trace.

(6) Keep the BST voltage path (BST, R_{BST}, C_2 and SW) as short as possible.

(7) Keep the IN and GND pads connected with large copper and use at least two layers for IN and GND trace to achieve better thermal performance. Also, add several Vias with 10mil_drill/18mil_copper_width close to the IN and GND pads to help on thermal dissipation.

(8) Four-layer layout is strongly recommended to achieve better thermal performance.



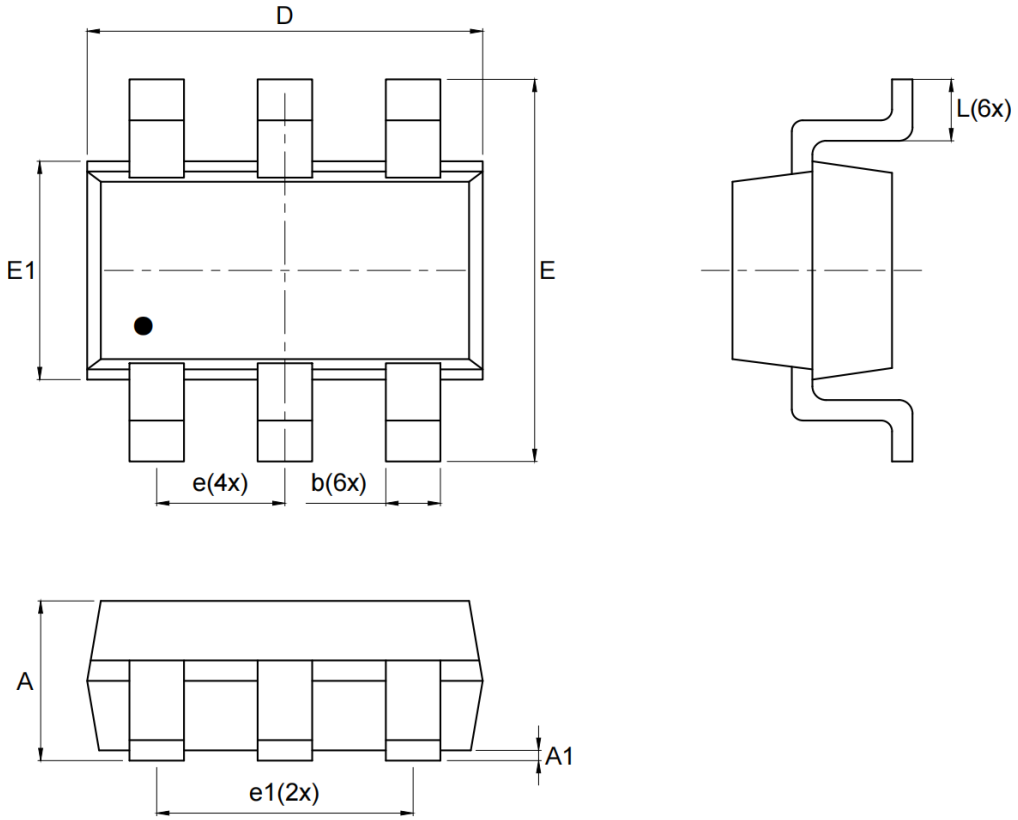


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PACKAGING INFORMATION

SOT23_6L Outline Dimensions

Unit: inches/mm

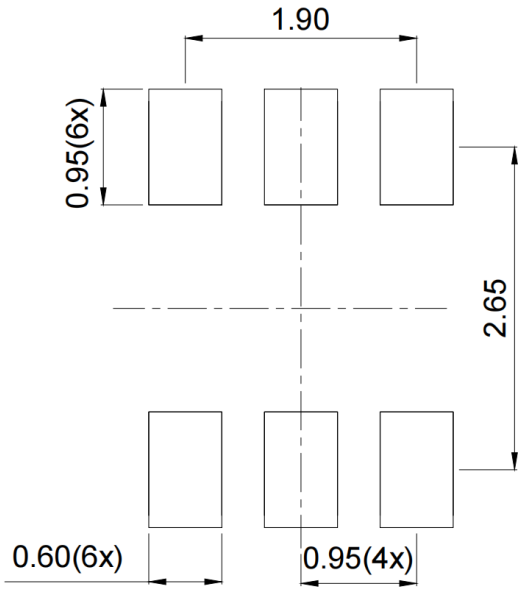


SYMBOLS	MILLIMETERS		INCHES	
	MIN.	MAX.	MIN.	MAX.
A	0.89	1.45	0.035	0.057
A1	0.00	0.15	0.000	0.006
b	0.30	0.50	0.012	0.020
D	2.70	3.10	0.106	0.122
E1	1.40	1.80	0.055	0.071
e	0.95 BSC		0.037 BSC	
e1	1.90 BSC		0.075 BSC	
E	2.60	3.00	0.102	0.118
L	0.30	0.60	0.012	0.024

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LAND PATTERN INFORMATION

SOT23_6L



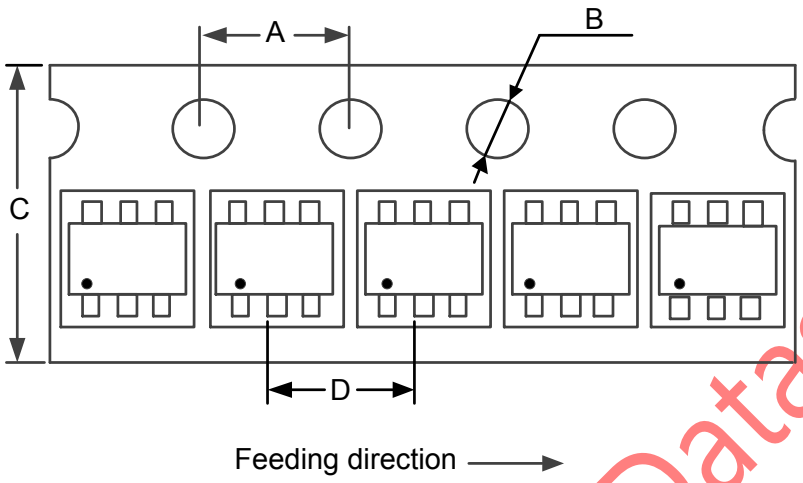
Note:

- Dimensions in mm.
- For reference only.

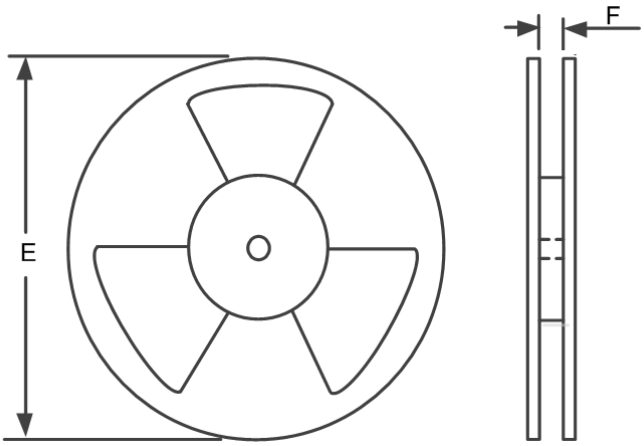
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CARRIER TAPE & REEL DIMENSIONS

1. Orientation / Carrier Tape Information:



2. Reel Information:



3. Dimension Details:

PKG Type	A	B	C	D	E	F	Q'ty/Reel
SOT23_6L	4.0 mm	1.5 mm	8.0 mm	4.0 mm	7 inches	9.0 mm	3,000

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REFLOW PROFILE

Classification Of IR Reflow Profile

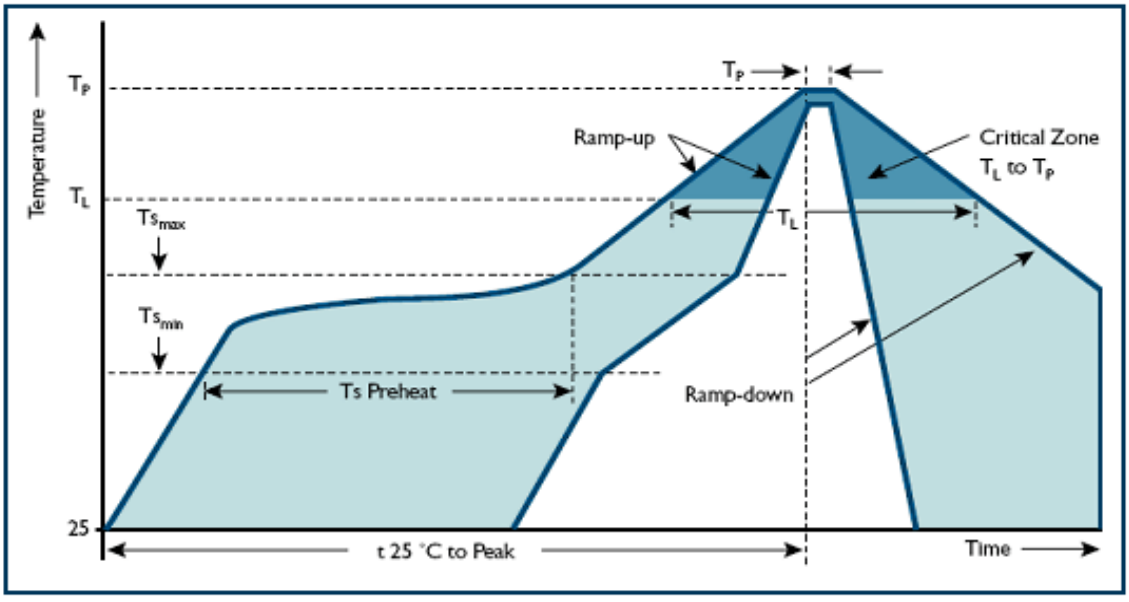
Reflow Profile	Green Assembly
Average Ramp-Up Rate (Ts_{min} to Tp)	1~2°C/second, 3°C/second max.
Preheat & Soak -Temperature Min(Ts_{min}) -Temperature Max(Ts_{max}) -Time($ts_{min}$to $tsts_{max}$)	 150°C 200°C 60~120 seconds
Time maintained above: -Temperature(T_L) -Time(t_L)	 217°C 60~150 seconds
Peak Temperature(Tp)	See Classification Temp intable1
Time within 5°C of actual Peak Temperature(tp)	30 seconds max.
Ramp-Down Rate	6°C/second max.
Time 25°C to Peak Temperature	8 minutes max.

* Tolerance for peak profile Temperature(T_p) is defined as a supplier minimum and a user maximum.
** Tolerance for time at peak profile temperature (t_p)is defined as a supplier minimum and a user maximum.

Table 1. Pb-free Process –Classification Temperatures (Tc)

Package Thickness	Volume mm^3 <350	Volume mm^3 350-2000	Volume mm^3 >2000
<1.6mm	260 °C	260 °C	260 °C
1.6mm–2.5mm	260 °C	250 °C	245 °C
2.5mm	250 °C	245 °C	245 °C

Note: For all temperature information, please refer to topside of the package, measured on the package body surface.



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